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Assignment # 3

EXAMPLE 2: Machine Selection - Complete 10-Year Analysis

Problem Statement

- **Machine A:** Economic life = 4 years, AETC = Rs. 2,780
- **Machine B:** Initial cost = Rs. 6,000, Year 1 maintenance = Rs. 1,500, Annual increment = Rs. 300
- **Interest rate:** 15% per annum
- **Question:** Find economic life and AETC of Machine B

Complete Calculation Table for Machine B (10 Years Analysis)

Year (n)	Maint e Cost (Rs.)	P/F @ 15%	PW of Mainten ance (Rs.)	Cumula tive PW of Mainte nance (Rs.)	Initia l Cost (Rs.)	Total PW Cost (Rs.)	CRF @ 15%	Annual Equival ent Total Cost (Rs.)
A	B	C	D (B×C)	ΣD	E	F (E+ΣD)	G	H (F×G)
1	1,500	0.8696	1,304.40	1,304.40	6,000	7,304.40	1.1500	8,400.0
2	1,800	0.7561	1,360.98	2,665.38	6,000	8,665.38	0.6151	5,328.7
3	2,100	0.6575	1,380.75	4,046.13	6,000	10,046.1	0.4380	4,420.1
4	2,400	0.5718	1,372.32	5,418.45	6,000	11,418.4	0.3503	3,999.9
5	2,700	0.4972	1,342.56	6,761.01	6,000	12,761.0	0.2983	3,805.0
6	3,000	0.4323	1,296.90	8,057.91	6,000	14,057.9	0.2642	3,714.3
7	3,300	0.3759	1,239.47	9,297.38	6,000	15,297.3	0.2404	3,677.5
8	3,600	0.3269	1,176.84	10,474.2	6,000	16,474.2	0.2229	3,669.8
9	3,900	0.2843	1,109.77	11,584.0	6,000	17,584.0	0.2096	3,685.62
10	4,200	0.2472	1,038.24	12,622.2	6,000	18,622.2	0.1993	3,710.98

Key Calculations Explained

1. Maintenance Cost (Column B)

- Year 1: Rs. 1,500
- Year 2: Rs. 1,500 + Rs. 300 = Rs. 1,800
- Year 3: Rs. 1,800 + Rs. 300 = Rs. 2,100
- And so on... (arithmetic progression with increment Rs. 300)

2. Present Value Factor (Column C)

$$PV Factor = \frac{1}{(1+i)^n} = \frac{1}{(1.15)^n}$$

Example: - Year 1: $\frac{1}{(1.15)^1} = 0.8696$ - Year 2: $\frac{1}{(1.15)^2} = 0.7561$ - Year 10: $\frac{1}{(1.15)^{10}} = 0.2472$

3. Present Worth of Maintenance (Column D)

$$PW = \text{Maintenance Cost} \times PV Factor$$

Example Year 2: $1,800 \times 0.7561 = \text{Rs. } 1,360.98$

4. Total PW Cost (Column F)

$$Total PW = Initial Cost + Cumulative PW of Maintenance$$

Example Year 4: $6,000 + 5,418.45 = \text{Rs. } 11,418.45$

5. Capital Recovery Factor (Column G)

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1}$$

Example for n=4:

$$CRF = \frac{0.15(1.15)^4}{(1.15)^4 - 1} = \frac{0.2624}{0.7490} = 0.3503$$

6. Annual Equivalent Total Cost (Column H)

$$AETC = Total PW Cost \times CRF$$

Example Year 4: $11,418.45 \times 0.3503 = \text{Rs. } 3,999.98 \approx \text{Rs. } 4,000$

7. Optimal Economic Life for Machine B

- **Minimum Annual Equivalent Total Cost occurs at Year 8: Rs. 3,669.86**

- This is the optimal time to replace Machine B

8. Conclusion

- Machine A's AETC of Rs. 2,780 is consistently lower than Machine B
- Machine B's minimum AETC (Rs. 3,669.86 at year 8) is still higher than Machine A
- **Total savings with Machine A:** Up to Rs. 890 per year (if comparing at 8 years)